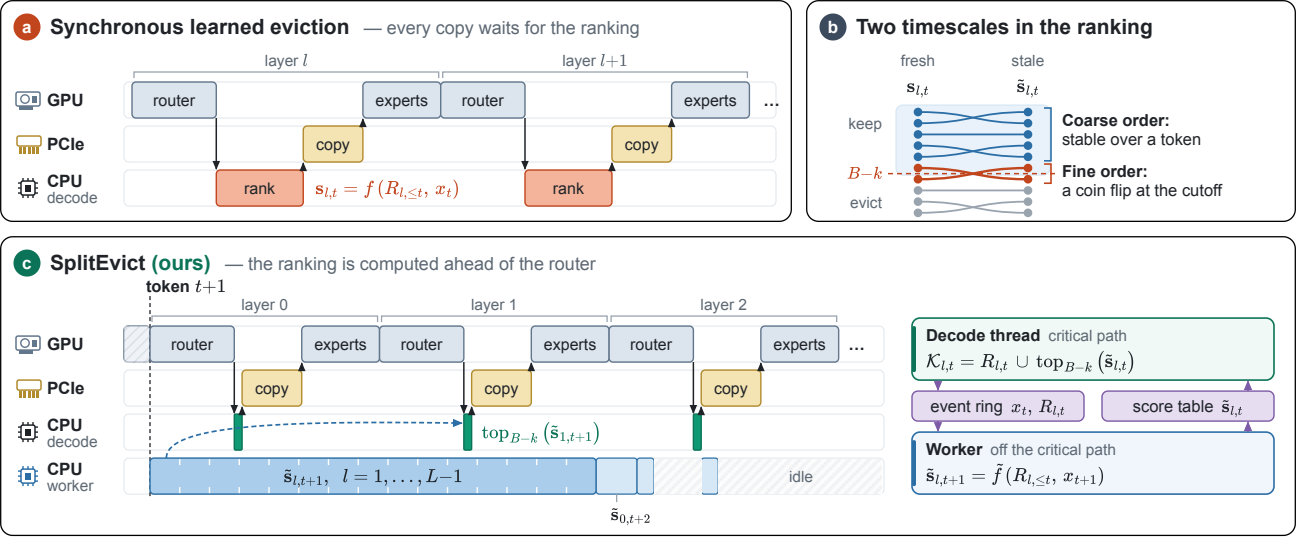

a Synchronous learned eviction — every copy waits for the ranking
layer l
layer $l+1$
GPU
PCIe
CPU decode
router
experts
router
experts
...
copy
copy
rank
$\mathbf{s}_{l,t} = f(R_{l,\le t}, x_t)$
rank
b Two timescales in the ranking
fresh
stale
$\mathbf{s}_{l,t}$
$\tilde{\mathbf{s}}_{l,t}$
keep
$B-k$
evict
Coarse order:
stable over a token
Fine order:
a coin flip at the cutoff
c SplitEvict (ours) — the ranking is computed ahead of the router
token $t+1$
layer 0
layer 1
layer 2
GPU
PCIe
CPU decode
CPU worker
router
experts
router
experts
router
experts
...
copy
copy
copy
$\mathrm{top}_{B-k}(\tilde{\mathbf{s}}_{1,t+1})$
$\tilde{\mathbf{s}}_{l,t+1},\ l = 1, \ldots, L-1$
idle
$\tilde{\mathbf{s}}_{0,t+2}$
Decode thread critical path
$\mathcal{K}_{l,t} = R_{l,t} \cup \mathrm{top}_{B-k}(\tilde{\mathbf{s}}_{l,t})$
event ring x_t, $R_{l,t}$
score table $\tilde{\mathbf{s}}_{l,t}$
Worker off the critical path
$\tilde{\mathbf{s}}_{l,t+1} = \tilde{f}(R_{l,\le t}, x_{t+1})$